

SALESPERSON PERFORMANCE ANALYSIS

Complete Project Report — Task 27

Data Analytics Track | Veda Technology

Tools: SQL • Excel • Power BI

TOTAL SALES	TOTAL PROFIT	PROFIT MARGIN	TOTAL ORDERS
\$4,358,061.68	\$172,970.99	3.97%	5,009
AVERAGE ORDER VALUE	SALESPERSONS	TERRITORIES	DATA ROWS
\$870.05	12	4	9,994

This report evaluates salesperson performance using multiple business KPIs rather than revenue alone. The analysis combines sales, profit, year-over-year growth and average order value, while retaining territory context.

Prepared deliverables: Excel workbook, SQL analysis, Power BI-ready dataset, DAX measures, dashboard preview, ranking, territory analysis and recommendations.

1. Executive Summary

The analysis covers 9,994 transaction rows, 5,009 distinct orders and 12 salespeople across 4 territories. Total sales are **\$4,358,061.68**, total profit is **\$172,970.99**, and overall profit margin is **3.97%**. The average order value is **\$870.05**.

Key findings

- Kabir Singh has the highest sales in the analysis at \$484,955.95.
- Diya Sharma has the highest total profit at \$21,753.01.
- Ishita Kapoor records the highest 2017-vs-2016 sales growth at 36.99%.
- Vihaan Kumar has the lowest total profit among the salespeople at \$7,730.94.
- Furniture — Tables is the lowest-profit sub-category at \$-27,042.32.

Why revenue alone is insufficient

Revenue indicates the value sold, but it does not show the cost of generating that revenue. A salesperson may have high sales but lower profitability because of discounts, product mix, returns or other commercial factors. Therefore this project evaluates several KPIs together.

2. Objectives

- Compare salespeople using Sales, Profit, Growth and Average Order Value.
- Create a transparent multi-KPI performance score.
- Compare performance with territory/region context.
- Identify profitable and loss-making product areas.
- Build reusable SQL, Excel and Power BI outputs.
- Provide practical recommendations for management and sales coaching.

3. Dataset & Data Preparation

The analysis uses a Superstore-compatible transactional dataset. Each row represents a sales line and contains order, customer, product, geography, sales, quantity, discount, profit and salesperson information.

Field Group	Examples
Order information	Order ID, Order Date, Ship Date, Ship Mode
Customer	Customer ID, Segment
Geography	Country, City, State, Postal Code, Region
Product	Product ID, Category, Sub-Category, Product Name
Measures	Sales, Quantity, Discount, Profit
Performance dimension	Salesperson

Data preparation: dates were parsed, year and profit-margin fields were derived, AOV was calculated using distinct orders, and KPI values were aggregated at salesperson, territory, year and sub-category levels.

4. Performance Methodology

To avoid ranking salespeople on a single metric, four KPIs were normalized to a 0–100 scale. The final score is a weighted composite:

Component	Weight
Sales Score	30%

Profit Score	30%
Growth Score	20%
AOV Score	20%
Total	100%

Growth is calculated as 2017 sales compared with 2016 sales. Min-max normalization is used so that metrics measured in different units can contribute to one composite score. The score is intended as an analytical framework, not as a substitute for managerial review of territory conditions and qualitative factors.

5. Salesperson Performance Ranking

Rank	Salesperson	Territory	Sales	Profit	Growth	AOV	Score
1	Diya Sharma	West	\$459,321	\$21,753	22.6%	\$926	88.8
2	Ishita Kapoor	East	\$443,681	\$20,974	37.0%	\$911	87.9
3	Kabir Singh	West	\$484,956	\$16,349	-21.0%	\$895	62.8
4	Rohan Gupta	East	\$417,484	\$13,483	-10.0%	\$894	52.2
5	Neha Joshi	Central	\$319,852	\$15,436	19.0%	\$872	51.3
6	Ananya Verma	East	\$428,176	\$17,514	-28.1%	\$800	44.6
7	Kavya Iyer	South	\$277,158	\$13,524	-14.4%	\$956	42.8
8	Aarav Mehta	West	\$435,414	\$14,550	-26.6%	\$822	42.2
9	Arjun Patel	Central	\$324,610	\$12,857	4.3%	\$856	39.9
10	Meera Nair	South	\$235,394	\$10,992	4.9%	\$826	22.0
11	Aditya Rao	South	\$222,761	\$7,809	-6.2%	\$913	21.4
12	Vihaan Kumar	Central	\$309,255	\$7,731	-25.8%	\$799	10.6

Interpretation: The ranking combines four dimensions. A salesperson with high sales but weak profitability or AOV may therefore not have the same composite score as a salesperson with a more balanced profile.

6. Territory Analysis

Territory	Sales	Profit	Orders	AOV	Margin	Sales Share
Central	\$953,717	\$36,024	1,133	\$842	3.8%	21.9%
East	\$1,289,341	\$51,970	1,489	\$866	4.0%	29.6%
South	\$735,312	\$32,325	819	\$898	4.4%	16.9%
West	\$1,379,691	\$52,652	1,568	\$880	3.8%	31.7%

Territory differences should be visible when comparing salespeople because market size, customer mix, demand, product mix, competitive intensity and discount practices can vary by geography. For this reason, the dashboard includes territory as a slicer and comparison dimension.

7. Year-wise Business Trend

Year	Sales	Profit	Orders	Sales Growth	Margin
2014	\$1,083,037	\$41,592	1,251.0	—	3.8%
2015	\$1,132,149	\$55,293	1,280.0	4.5%	4.9%
2016	\$1,101,698	\$34,318	1,265.0	-2.7%	3.1%
2017	\$1,041,178	\$41,768	1,213.0	-5.5%	4.0%

8. Product & Sub-Category Profitability

Category	Sub-Category	Sales	Profit	Margin	Avg Discount
Furniture	Tables	\$328,080	\$-27,042	-8.2%	16.5%
Furniture	Chairs	\$359,961	\$-4,194	-1.2%	17.0%
Furniture	Furnishings	\$331,918	\$-925	-0.3%	17.9%
Furniture	Bookcases	\$367,157	\$1,496	0.4%	16.6%
Office Supplies	Envelopes	\$86,444	\$2,279	2.6%	18.8%
Office Supplies	Appliances	\$81,456	\$2,741	3.4%	17.8%
Office Supplies	Art	\$91,672	\$3,020	3.3%	18.0%
Office Supplies	Supplies	\$89,628	\$3,694	4.1%	18.7%
Office Supplies	Fasteners	\$77,031	\$4,044	5.2%	16.9%
Office Supplies	Labels	\$81,147	\$4,185	5.2%	17.3%
Office Supplies	Storage	\$86,292	\$4,465	5.2%	15.9%
Office Supplies	Paper	\$93,303	\$4,831	5.2%	17.6%
Office Supplies	Binders	\$83,174	\$5,183	6.2%	16.9%
Technology	Copiers	\$523,823	\$32,068	6.1%	17.5%
Technology	Accessories	\$581,179	\$32,224	5.5%	18.7%
Technology	Phones	\$577,050	\$42,279	7.3%	17.2%
Technology	Machines	\$518,748	\$62,624	12.1%	16.7%

Loss-making sub-categories deserve attention because increasing their volume without addressing pricing, discounts or costs can reduce overall profitability. The SQL analysis also groups transactions by discount band to support discount-policy review.

9. Power BI Dashboard Design

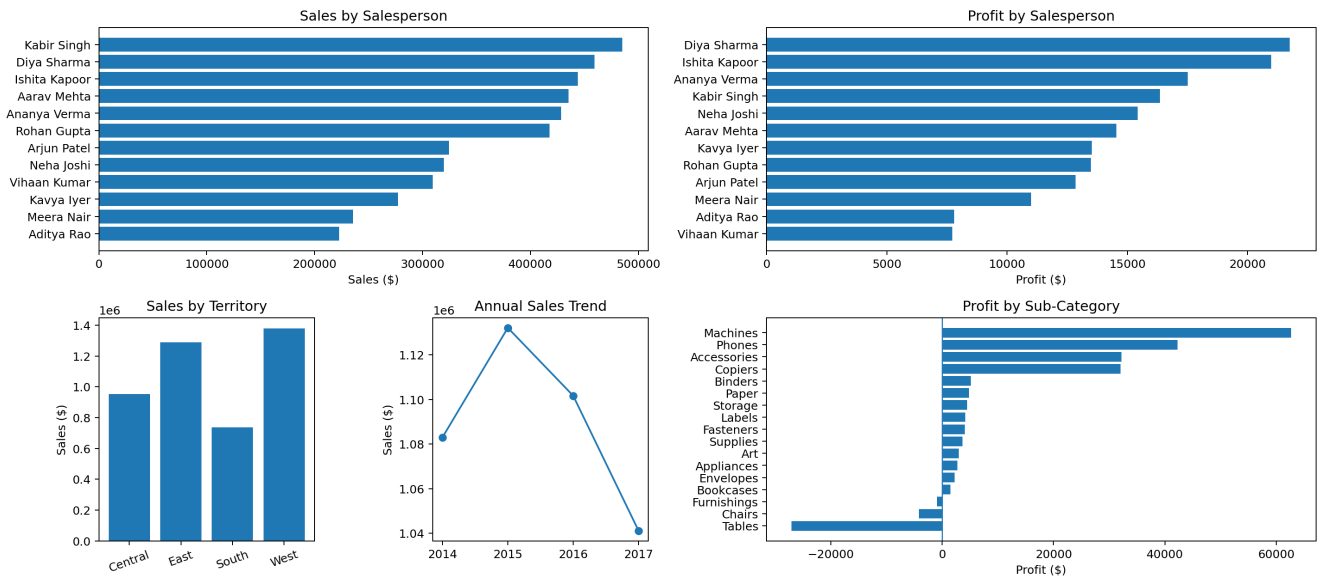
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Total Sales
\$4,358,062

Total Profit
\$172,971

Profit Margin
3.97%

Orders
5,009



- KPI cards: Total Sales, Total Profit, Profit Margin, Orders and AOV.
- Sales and Profit by Salesperson.
- Composite Performance Score by Salesperson.
- Annual/monthly sales trend.
- Profit by sub-category.
- Sales by territory.
- Sales vs Profit scatter plot.
- Slicers: Year, Territory/Region, Category, Segment and Salesperson.

10. SQL & Excel Deliverables

The SQL script contains queries for overall KPIs, salesperson performance, 2016–2017 growth, territory comparison, category/sub-category profitability, discount bands, loss-making salesperson lines and top products.

The Excel workbook contains separate sheets for the raw/clean data, KPI Summary, Salesperson Ranking, Territory Analysis, Year Analysis and Subcategory Analysis. Charts are included for salesperson profit and annual sales trend.

11. Business Recommendations

- Use a multi-KPI scorecard for periodic salesperson reviews instead of relying only on revenue.
- Review salesperson performance within territory context before drawing conclusions from absolute sales.
- Investigate loss-making sub-categories and high-discount transactions for pricing and margin improvement.
- Monitor growth together with profit so revenue expansion does not hide margin deterioration.
- Use AOV and order count to identify cross-selling and basket-size opportunities.
- Refresh the Power BI dashboard regularly and create exception flags for negative profit and declining growth.
- Combine quantitative results with qualitative factors such as account difficulty, market conditions and customer mix.

12. Interview Questions & Answers

Why is revenue alone insufficient?

Revenue measures the value sold but does not show costs or profitability. A salesperson can have high revenue but lower profit because of discounts, product mix or other commercial factors.

How can territory affect comparison?

Territories can differ in market size, customer mix, demand, competition and discount levels. Therefore salesperson KPIs should be interpreted with territory context.

Why use multiple KPIs?

Multiple KPIs capture volume, profitability, growth and order economics. This reduces dependence on one metric.

How is AOV calculated?

AOV is Total Sales divided by the number of distinct orders.

How is growth calculated?

For this project, growth is calculated as $(2017 \text{ Sales} - 2016 \text{ Sales}) \div 2016 \text{ Sales} \times 100$.

Why normalize KPIs?

Sales, profit, growth and AOV have different units and scales. Normalization places them on a common 0–100 scale before weighting.

13. Conclusion

The completed analysis demonstrates how SQL, Excel and Power BI can be combined to evaluate salesperson performance in a structured way. The project moves beyond a simple sales leaderboard by incorporating profitability, growth, AOV and territory context. The resulting dashboard and supporting files provide a reusable framework for management reporting and interview discussion.

Submission package: Clean dataset • Power BI dataset • Excel workbook • SQL script • DAX setup • Dashboard preview • Ranking CSV • Territory CSV • Recommendations CSV • This complete PDF report.